

Lab Task on 2D Array

1. Create a Program that can Generate any odd sized magic square.
 - a. Take user input for Dimension Size and should be an Odd Number.
 - b. Generate and store the Magic Square in a 2D dynamic array.
 - c. Sum any one side or diagonals and show the summation result.

```
Dimension of Magic Square Matrix : 3
8      ①      6
3      5      7
4      9      2
Sum of Any Side or Diagonals: 15
```

```
Dimension of Magic Square Matrix : 5
17     24     ①     8     15
23     5      7     14    16
4      6      13    20    22
10     12     19    21    3
11     18     25    2     9
Sum of Any Side or Diagonals: 65
```

Rules for Generating Magic Square:

1. Initialize (n x n) 2D array with 0.
2. For N Dimension, Number will start from **1 to N²** (**Example: if n=3, Elements are 1 to 9**) and a number can be used only once.
3. Start by assigning Element **1** to index **(n/2, n/2)** and consider it as the starting cell.
4. Check Upper-Right diagonal cell (**Note: index cannot go out of bound in any axis, it must circulate back to that axis**)
 - 4.1. if it is Zero assign the next element on that Upper-Right diagonal cell.
else assign the next element below current cell.
 - 4.2. Repeat step 4, until all cells are filled with unique numbers.